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Measurement of local impedance by an orbit bump method

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Abstract

A new method of local transverse coupling impedance measurement has been developed to estimate the impedance distribution along a storage ring. The method is based on measurement of the closed orbit deviation caused by the wake field generated at the impedance location by the off-axis beam. This method provides better accuracy than the known method based on betatron phase advance measurement, because the beam position monitors operate in narrow-band (orbit) mode, but not in wide-band (turn-by-turn) mode used for phase advance measurement. The physical principles and limitations of the method are described using both analytical formulae and computer simulation.

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1. Introduction

For many modern particle accelerators, the beam–environment interaction is a matter of interest. This interaction may cause various beam instabilities, which can limit beam intensity and deteriorate beam quality. The coupling impedance approach is widely used to study the beam–environment interaction in frequency domain. Knowledge of the coupling impedance is required in analytical or numerical estimations of the instability characteristics. Impedance calculation for a complex vacuum chamber is quite a cumbersome problem. In many cases, measurement of the impedance could be useful for the

understanding of various collective effects observed.

There have been advanced methods to measure the total impedance of a storage ring [1]. However, measurement of the impedance distribution is a more complicated problem. The longitudinal loss factor distribution can be obtained by measurement of horizontal closed orbit deviation versus beam current [2]. Measurement of the betatron phase advance allows to find the distribution of the reactive transverse impedance [2]. These methods work rather well if the impedance is located in few vacuum chamber sections, such as RF cavities, etc. At the electron–positron collider VEPP-4M [3], total impedance results from as many as about 50 discontinuities of the vacuum chamber. Complicated impedance distributions are also typical for synchrotron radiation sources with the vacuum chambers containing quite a

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number of narrow-gap insertions. Our attempts to apply the known methods [2] for impedance distribution measurement at the VEPP-4M collider failed, because each discontinuity gave rise to too small an effect within the r.m.s. error of the betatron phase advance measurement.

Then, we have developed another, more sensitive method of local transverse impedance measurement, based on closed orbit measurements. In 1998, this method was successfully used for measurement of impedance distribution at the VEPP-4M electron–positron collider [4].

2. Physical principles

The method proposed is based on the fact that an off-axis bunch passing through the vacuum chamber section with a non-zero transverse impedance is deflected by the wake field induced in the section by the bunch itself.

We consider two ultra-relativistic particles moving along the longitudinal axis in an axially symmetric environment. In cylindrical coordinates (r, θ, s) , the first, leading particle with a transverse offset $(r = a, \theta = 0)$ can be described as a driving current $\vec{j}$ decomposed in terms of multipole moments [5]

$$\begin{aligned} \vec{j} &= \vec{v} \frac{q}{a} \delta(\theta) \delta(r - a) \delta(s - vt) \\ &= \vec{v} \sum_{m=0}^{\infty} \frac{I_m \cos m\theta}{\pi a^{m+1} (1 + \delta_{m0})} \delta(r - a) \delta(s - vt) \end{aligned} \quad (1)$$

where q is the particle charge, $\vec{v}$ is the velocity in the s -direction, and $I_m = qa^m$ is the m th multipole moment. The wake force $\vec{F} = e(\vec{E} + \vec{v} \times \vec{B})$ of the electromagnetic field generated by the driving current $\vec{j}$ acts on the trailing particle with the elementary charge e . The wake potential is defined as the wake force $\vec{F}$ integrated along the s -direction over the structure length L , holding the distance z between the particles fixed. The transverse wake potential is [5]

$$\begin{aligned} \int_{-L/2}^{L/2} \vec{F}_{\perp} ds &= -eI_m W_m(z) m r^{m-1} \\ &\times (\hat{r} \cos m\theta - \hat{\theta} \sin m\theta) \end{aligned} \quad (2)$$

where W_m is the wake function for the moment m , (r, θ) are the coordinates of the trailing particle ($\hat{r}, \hat{\theta}$ are the unit vectors in the r and θ directions, accordingly). When the leading particle offset a is much less than the characteristic radius of the structure, the dipole moment $m = 1$ dominates, and Eq. (2) reduces to

$$\int_{-L/2}^{L/2} \vec{F}_{\perp} ds = -eI_1 W_1(z) a \hat{r} \quad (3)$$

where the trajectory of the trailing particle is taken to be the same $(r = a, \theta = 0)$ as the trajectory of the leading particle.

Thus, if the distance between the particles z is short enough (of the same order as the characteristic size of the structure), the trailing particle experiences a transverse kick in the r -direction:

$$\Delta r' = \frac{-eI_1 W_1(z) a}{E/c} \quad (4)$$

where E is the particle energy.

For a thin bunch with the longitudinal particle distribution ρ , the driving current is

$$\vec{j} = \vec{v} \frac{q}{a} \delta(\theta) \delta(r - a) \rho(s). \quad (5)$$

Here q is the bunch charge, and $(r = a, \theta = 0)$ is the transverse offset of the bunch. Taking the dipole moment I_1 of Eq. (5), by integration Eq. (4) over s and z , one can obtain the transverse kick of the bunch centroid [5]:

$$\Delta r' = \frac{q}{E/e} k_{\perp} a. \quad (6)$$

The transverse kick factor $k_{\perp}$ is a function of the dipole wake function W_1 (called also the transverse wake function $W_{\perp}$), and of the longitudinal particle distribution ρ :

$$k_{\perp} = \int \int \rho(s) \rho(s - z) W_1(z) ds dz \quad (7)$$

or, in the frequency domain,

$$k_{\perp} = \frac{1}{\pi} \int_0^{\infty} \Im Z_{\perp}(\omega) |\tilde{\rho}(\omega)|^2 d\omega \quad (8)$$

where $\Im Z_{\perp}(\omega)$ is the dipole transverse impedance of the structure, and $|\tilde{\rho}(\omega)|^2$ is the power spectral density of the spatial particle distribution $\rho(s)$. Up to a constant normalization factor, the kick factor

$k_{\perp}$ is equal to the effective transverse impedance [5] of the structure.

For a Gaussian bunch

$$\rho(s) = \frac{q}{\sqrt{2\pi}\sigma_s} \exp\left(-\frac{s^2}{2\sigma_s^2}\right) \quad (9)$$

the transverse dipole kick is

$$\Delta r' = \frac{I_p}{E/e} \Im Z_{\perp} a. \quad (10)$$

Here I_p is the peak bunch current defined for Gaussian bunch as

$$I_p = \frac{\sqrt{2\pi}R}{\sigma_s} I_b \quad (11)$$

where $I_b = qf_{\text{rev}}$ is the average bunch current, and R is the average machine radius.

Making the plane $\theta = 0$ agreed with the horizontal x - or vertical y -plane, kick (10) can be re-written for $\Delta x'$ and $\Delta y'$ accordingly. So, if a short enough Π -shape distortion (bump) of the closed orbit is created at the impedance location s_0 , then, due to the kick, an orbit deviation beyond the bump appears while increasing the bunch peak current. This orbit deviation has the form of standing betatron waves. For the y -plane, with the bump height $a = y_0$, it is

$$\Delta y(s) = \frac{\Delta I_p y_0}{2(E/e) \sin \pi \nu} \Im Z_{\perp} \sqrt{\beta(s_0)\beta(s)} \times \cos(\Delta\phi - \pi\nu) \quad (12)$$

where ν is the betatron tune, $\Delta\phi = \phi(s) - \phi(s_0)$ is the betatron phase advance, β is the amplitude betatron function, and ΔI_p is a change in the peak bunch current. This orbit wave can be measured using a beam position monitor (BPM) system, and the wave amplitude yields information about reactive component of the dipole transverse impedance at the bump location.

3. Applicability and limitations

To see the practical domain of the method applicability, let us estimate an order of value of orbit deviation (12). There is a known formula for a current-dependent coherent shift of a betatron tune ν caused by the reactive dipole transverse

impedance $\Im Z_{\perp}$:

$$\frac{\Delta\nu}{\Delta I_p} = -\frac{1}{8\pi} \frac{\Im Z_{\perp} \beta}{E/e} \quad (13)$$

where β is the beta function at the impedance location. For the case of a total impedance concentrated in a single location, the wave amplitude can easily be found from Eqs. (12) and (13)

$$\Delta y \approx 4\pi \Delta\nu y_0. \quad (14)$$

For the VEPP-4M storage ring, a measured coherent tune shift at the 1.8 GeV energy is typically $(\Delta\nu/\Delta I_b) = -1.5 \text{ A}^{-1}$. So, for a change of the average beam current $\Delta I_b = 0.01 \text{ A}$ ($\Delta I_p = 36.5 \text{ A}$ when $\sigma_s = 4 \text{ cm}$) and for a bump height $y_0 = 5 \text{ mm}$, the orbit deviation would be about $\Delta y \approx 1 \text{ mm}$. The effect has been quite noticeable. One can expect that the orbit deviation can be observed and measured even with a partial, local impedance.

The betatron wave (12) is derived for the case of a short bump, with amplitude $\beta(s_0)$ and phase $\phi(s_0)$ betatron functions assumed constant through its length. In practice, it is not always possible to create a bump short enough using the orbit correctors available. For a long bump, the orbit deviation can be derived from a generalized equation for extended kick, and it reads as:

$$\Delta y(s) = \frac{\Delta I_p y_0 \sqrt{\beta(s)}}{2(E/e) \sin \pi \nu} \int_{s_0}^{s_0+\Delta s} \Im Z_{\perp}(s') \sqrt{\beta(s')} \times \cos(\Delta\phi - \pi\nu) ds' \quad (15)$$

here $\Delta\phi = \phi(s) - \phi(s')$. The orbit deviation (15) is proportional to the impedance β -weighted over the bump length Δs .

With a longer bump, for illustration of situations that may have place, let us consider some special impedance distribution when there are two equal local impedances within the bump with a betatron phase advance between them close to π . The orbit deviation manifests itself within the bump only, outside the bump it comes to zero. In this situation, a small number of the orbit BPMs is used, and the measurement accuracy deteriorates.

Note that expressions (12) and (15) are derived with that assumption that $\sigma_{\perp} \ll y_0$ where $\sigma_{\perp}$ is the

transverse beam size, y_0 is the bump height. If this condition is not satisfied then the kick factor should be used as Eq. (7) averaged over the bunch cross-section as well as over the bunch length.

Let us consider the bump method in application to structures of some characteristic cross-sections. First, return to the axially symmetric case and suppose that the bump height is large. The total kick of the bunch centroid is a sum of the kicks from each m -component of the wake potential (2). So, some algebraic sum of the $\Im Z_m$ impedances can be determined from Eq. (12). Thus, in general, the dipole impedance $\Im Z_{\perp}$ can only be determined by the method, under the condition that the bump height is considerably less than the transverse characteristic size of the structure. Note, that the wake potential (2) scales with the transverse offset r as mr^{2m-1} , so the $m > 1$ components are quite small in the vicinity of the beam axis.

For structures symmetric about the cartesian coordinate planes, in the coordinate center vicinity, the wake potential (2) can approximately be represented as two individual dipole terms for both horizontal (x) and vertical (y) directions. Thus, in the vicinity, the case is similar to the axially symmetric case, and for structures like elliptic or rectangular collimators, step or tapered junctions, etc., it is also possible to determine the $\Im Z_{\perp x}$, $\Im Z_{\perp y}$ dipole components provided that the bump height is considerably less than the characteristic size of the structure. For an off-axis beam, the cross terms in the wake field expansion arise, and a notable part of the transverse kick is determined by the quadrupole ($m = 2$) harmonic.

Finally, we consider structures symmetrical about only one of the Cartesian coordinate planes, like beam probes, scrapers, pin-type synchrotron radiation collectors, etc. A conclusion about the impedance of such a structure can be made only if the wake field two-dimensional topography is measured in sufficient detail.

We have carried out a wake field computer calculation for a vertical probe installed on the VEPP-4M storage ring. Its face is placed at -6 mm from the beam axis (see Fig. 1). The calculated distribution of the horizontal magnetic field (which deflects the bunch vertically) along the vertical axis is shown in Fig. 2. The bunch charge

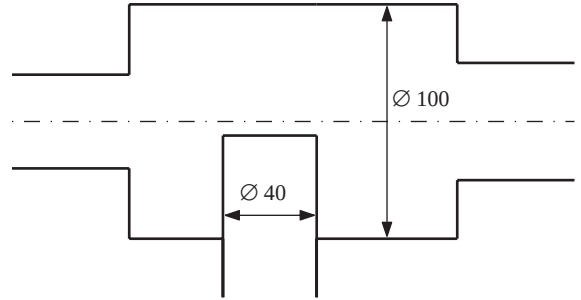


Fig. 1. Probe layout.

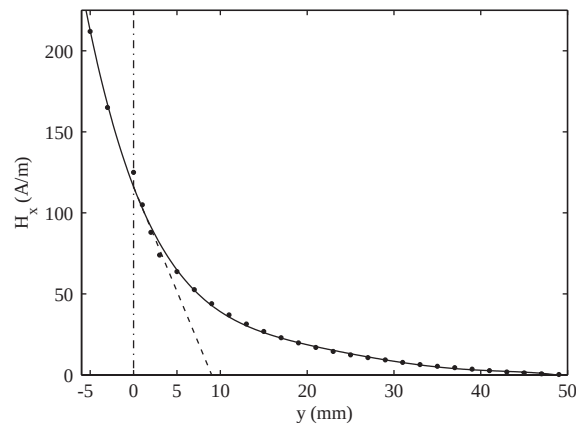


Fig. 2. Calculated wake field distribution.

is 6 nC, its position is in the coordinate center, and the bunch length is $\sigma_s = 4$ cm.

One can see that this transverse wake field is non-zero everywhere including the center. The field is non-linear as well. One can expect that the transverse wake potential is also non-zero at the center, contrary to the previous case, where the potential comes to zero as it does in the axially symmetric structure. In the vicinity of the center, the potential can be expanded in the series

$$\int_{-L/2}^{L/2} \vec{F}_{\perp} ds = A_0 + A_1 y + A_2 y^2 + \dots \quad (16)$$

where y is the coordinate of the trailing particle, A_0 is the value of the wake potential in the center, $A_1, A_2, \dots$ are coefficients. Neglecting the non-linearity, one can say that the particle gets a kick from the first term as from a dipole corrector, and a defocusing kick from the second term, which is similar to the dipole term (3) in the symmetric

case. One can find the impedance of interest, corresponding to the dipole term, with the bump method if to measure (12) for two bumps of opposite polarities and take half of difference of these two orbit deviations. Note that the first kick is much stronger than the second one. Note also that in principle, the orbit deviation from the first term can be measured separately, with a bunch in the center and a change of the peak current up to a nominal value from some low value, that one when the kick occurred is negligible.

It may be acceptable from the accuracy point of view to find orbit deviation (12) corresponding to the dipole impedance, using a measured value of the deviation caused by the term A_0 , and a ratio A_1/A_0 of the numerically calculated values.

The case of horizontal impedance measurement here is close to the case of structure symmetric about the both cartesian coordinate planes, with that difference that the measured impedance depends now on beam displacement along the transverse coordinate.

The method can be modified for the case when the vacuum chamber section has an element which can be moved (probe, scraper, etc.). With a movable element, no bump and no change of the bunch current are necessary. The orbit deviation appears when the element is moved from the periphery of the section close to the beam. When measured by this method the vertical effective impedance of the probe of the VEPP-4M is about 32 k Ω /m, in the quasi-linear field region of ± 5 mm (see Fig. 2), the wake field values, calculated from the impedance measured, correspond rather well with the simulation result.

One of the authors has also applied the modified bump method at the ELETTRA synchrotron light source. Transverse profile of the wake field of a rod-type scraper was measured with an accuracy exceeding the accuracy of the coherent tune shift method [6]. The scraper contribution to the total impedance has been measured too.

4. Measurement technique and accuracy

The measurement procedure has been designed for reducing the effect of a systematic beam

intensity-dependent BPM error which may make the impedance measurement unreliable. In the procedure, the error is also measured and then subtracted from Eq. (12). First of all, after correcting the initial orbit to zero, two reference orbits y_{01} and y_{02} are measured at high and low values of beam current. Then, after creating the orbit bump, again two orbits y_1 and y_2 are measured at the same current values. In the following four-orbit combination for Δy of (12)

$$\Delta y = (y_2 - y_1) - (y_{02} - y_{01}) \quad (17)$$

the systematic error has been eliminated, as well as the bump itself.

The accuracy depends also on the BPM noise resolution δy :

$$\frac{\delta Z_{\perp}}{Z_{\perp}} = 2 \frac{\delta y}{\Delta y} \frac{1}{\sqrt{N}} \quad (18)$$

where N is the number of BPMs, and factor 2 is due to combining four individual orbit measurements in Eq. (17). The advantage of this method, in comparison with the method based on betatron phase advance measurement, is that is the increase of accuracy $\sqrt{N}$ times, because all the $N \gg 1$ BPMs are used for the impedance measurement.

A vertical orbit bump shown in Fig. 3 was created at one of the high-impedance sections of the VEPP-4M storage ring. The section contains 3 of 5 RF cavities, 3 vertical strip-lines, and 5 vertical step irregularities in the vacuum chamber. The measured beta-weighted impedance of the section is 2.1 M Ω which is about 10% of the total reactive transverse impedance of the ring. Fig. 4 shows both the horizontal and vertical orbit combinations (17) measured by the BPMs. A change of the peak current is $\Delta I_p = 20$ A. The solid line represents the orbit wave calculated with (12). As one can see, the measured vertical orbit agrees quite well with the wave (12), while the horizontal one stays within the measurement resolution, represented by the error bars. The r.m.s. deviation of the vertical measured data from the calculated one is 0.016 mm, which is comparable with the BPM noise resolution $\delta y = 0.021$ mm. This fact means that all the assumptions implied in Eq. (12) are valid, and the method is used properly.

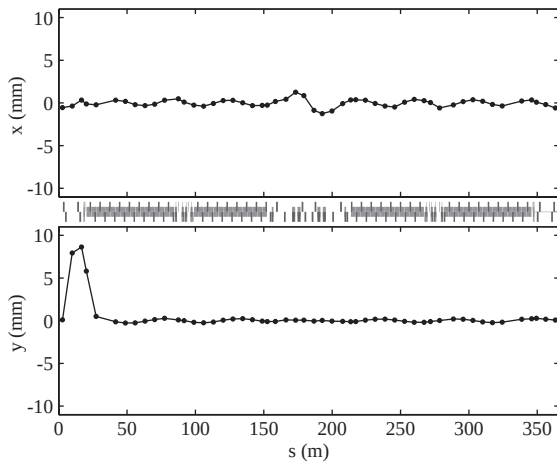


Fig. 3. Vertical orbit bump.

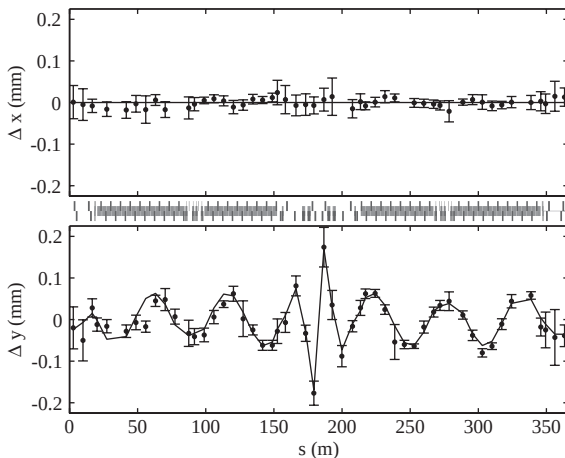


Fig. 4. Vertical orbit deviation caused by a wake field.

5. Conclusions

The bump method used carefully can give information about a local coupling impedance. This method can provide better accuracy than the known one based on the betatron phase advance measurement, because beam position monitors operate in narrow-band (orbit) mode, but not in

wide-band (turn-by-turn) mode, and the BPM noise resolution is proportional to the square root of the bandwidth. Another advantage of the method is use of all the BPMs for the local impedance evaluation, which increase the accuracy $\sqrt{N}$ times, where N is the number of BPMs. In general, the bump method provides a beam-based measurement of two-dimensional topology of a wake field which act on a bunch at the orbit bump location. Frequency dependence of the impedance can also be measured if bunch length is varied.

Taking into account the limitations described above, the bump method was successfully used for measurement of impedance distribution along the VEPP-4M storage ring [4]. Using a modified bump method, the transverse profile of the wake field of a rod-type scraper and the scraper contribution to the total impedance has been measured at ELETTRA [6].

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